

## REPRODUCTION OF THE SPINY LOBSTER *PANULIRUS GUTTATUS* (DECAPODA: PALINURIDAE) ON THE CARIBBEAN COAST OF MEXICO

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### ABSTRACT

The reproductive biology of *Panulirus guttatus* from northern Quintana Roo, Mexico, was studied as part of a large research program on the biology and ecology of this poorly studied species. Lobsters were caught with baited traps deployed on the coral reef throughout 17 months (May 1988–September 1989). A total of 870 lobsters (558 males and 312 females) were obtained. The strongly biased sex ratio may have been an artifact of the sampling method, but differential movements of males and females related to reproductive activity were not excluded as possible causes. Ovigerous females (40% of total females, i.e., 125) occurred throughout the year, but were more prevalent during the spring of 1988, and the spring and summer of 1989. The size range of ovigerous females was 40.0–73.5 mm in carapace length (CL). Brood size was determined for the 125 ovigerous females plus an additional 41 sampled during 1987. After excluding the data of females with eggs about to hatch (of smaller sizes than eggs in previous stages), seasonal log brood size–log CL relationships were examined. No significant differences were found either seasonally or inter-annually, indicating that the average brood size of females of equal CL remained fairly similar throughout the study period. Females between 55 and 65 mm CL had the highest indices of reproductive potential, and accounted for 54% of the total egg production of the population. These results were compared with others obtained from populations of *P. guttatus* in Florida and other parts of the Caribbean.

*Panulirus guttatus* (Latreille, 1804) is a small tropical lobster occurring throughout the Caribbean, the Bahamas, and Bermuda. It cooccurs with *P. argus*, the common Caribbean spiny lobster, but, in contrast to the latter species which occupies a wide range of habitats, *P. guttatus* is restricted to the coral reef habitat and is not found in sea grass or sandy areas far from the coral reef. The puerulus of this species possibly settles directly onto the coral reef habitat and remains in it throughout its benthic life (Briones-Fourzán and McWilliam, 1997).

In Mexico, *P. guttatus* occurs throughout the eastern coast of the Yucatan Peninsula, and accounts for 6% of the spiny lobster catch in the northern part of this coast (Padilla-Ramos and Briones-Fourzán, 1997), where it is caught mainly by SCUBA or skin-diving.

Owing to its small size and low overall abundance, *P. guttatus* has minor importance as a fishing resource and is a poorly studied species. The reproductive biology of *P. guttatus* has been studied in Florida (Chitty, 1973) and Martinique (Marfin, 1978).

As part of a large research program conducted by the National University of Mexico on the biology of spiny lobsters from the

Mexican Caribbean, the reproductive biology of *P. guttatus* and *P. argus* were concurrently studied. Results on the latter species were published elsewhere (Fonseca-Larios and Briones-Fourzán, in press). In this paper, we present results on the sex ratios, reproductive seasons, size at first maturity, and estimates of brood size for *P. guttatus* from northern Quintana Roo, on the Caribbean coast of Mexico.

### MATERIALS AND METHODS

**Study Area.**—Lobsters were caught on the coral reef 10 km north of Puerto Morelos (20°54'N, 86°54'W), on the Mexican Caribbean coast. This barrier coral reef runs approximately parallel to the coastline, at a distance of approximately 500 m. Scleractinian corals, dominated by *Acropora palmata* (Lamarck) (see Jordán-Dahlgren, 1979), form the main structure of the reef, which constitutes the principal habitat for *Panulirus guttatus* in the study area (Carrasco-Zanini, 1985; Briones-Fourzán, 1991). The lagoon between the coast and the reef is covered with sea grasses, mainly *Thalassia testudinum* Banks ex König, and a vast array of macroalgae. The depth of sampling varied from <1 to 2.5 m, and water temperature during the sampling period ranged from 25.0–31.2° C.

**Lobster Sampling.**—Lobsters were caught with rectangular, double parlor collapsible traps, measuring 82 cm long × 51 cm wide × 30 cm high. The base and lid of the trap were made of steel, and the body of the trap was 4.5-cm mesh netting. Pieces of cow hide, placed in the

innermost parlor, were used as bait. Traps were installed under ledges and near crevices on the leeward side of the reef, and were checked early in the morning for 15 days each month, from the first to the third quarter moon phases, which is the time of maximum activity for *P. guttatus* (see Caillouet *et al.*, 1971; Carrasco-Zanini, 1985). Sampling was conducted from May to November 1987, when only ovigerous females were retained, and from May 1988 to September 1989, when all lobsters caught were measured and examined.

Carapace length (CL), taken from between the supra-orbital horns to the posterior edge of the carapace, was measured with calipers ( $\pm 0.1$  mm). Egg-bearing females were taken to the laboratory. The color of the egg mass was recorded as an indicator of the degree of development of the embryos: bright orange (recently oviposited), dark orange (intermediate development), and brown (about to hatch) (Fonseca-Larios and Briones-Fourzán, in press). Although the eggs were not measured, we used the average number of eggs in 0.1 g of egg mass (EM) as an index of the relative egg mass; the smaller the mean number of eggs in 0.1 g EM, the larger the mass of the eggs.

**Estimation of Brood Size.**—The pleopods of ovigerous females, bearing the egg mass, were cut from the abdomen and fixed in 70% ethanol. The eggs were later removed from the pleopods, placed on a mesh sieve for several hours to eliminate the excess fixative, and oven-dried at 45°C for 48 h (Morgan, 1972; Gracia, 1985). The eggs were then separated, sieved, thoroughly cleaned of any remaining debris, and weighed ( $\pm 0.001$  g). Three 0.01-g subsamples were separated and manually counted under a magnifying glass. The average number of eggs in the 3 subsamples was then multiplied by the total weight of the egg mass to obtain the total number of eggs for a particular individual.

For seasonal comparisons, data from May 1988 onward were grouped as follows: spring 1988 (May), autumn 1988 (September–November 1988), winter 1989 (December 1988–February 1989), spring 1989 (March–May 1989), and summer 1989 (June–August 1989). Summer 1988 was not included in seasonal comparisons, because only 7 ovigerous females were collected during this season. Regression equations were fitted to the seasonal data sets of log brood size (BS) versus log CL (Chubb, 1994). To test for the similarity in seasonal regressions, an analysis of covariance of the regression coefficients of autumn 1988, and winter, spring, and summer 1989 was performed (Zar, 1984). Regression coefficients for spring 1988 and spring 1989 were also compared to investigate whether the brood size was similar during the same season for two consecutive years.

Data from 1987 were not broken down by season, due to the small overall number of lobsters sampled during that year. However, once the seasonal data of 1988–1989 were compared, the regression coefficients of the whole samples of 1987 and 1988–1989 were compared to test the hypothesis of similarity in fecundity between 2 years.

**Index of Reproductive Potential.**—The index of reproductive potential (IRP) devised by Kanciruk and Herrin-kind (1976) determines the size classes of breeding females which contribute most to egg production in the population.  $IRP = (A \times B \times C)/D$ , where A = proportion of females in a particular size class per total number of females in the sample, B = proportion of ovigerous females in the size class, C = mean brood size in the size class,

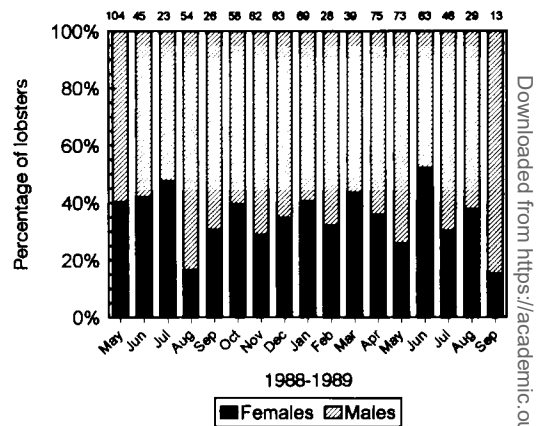


Fig. 1. Percentage of male and female *Panulirus guttatus* in monthly samples from May 1988 to September 1989. Numbers above bars indicate sample size.

and D = a constant, which is computed by setting the IRP of 1 of the size classes to 100 as the standard. To construct the IRP, we used the CL data of the females obtained during 1988 and 1989, arranged by 5-mm size classes.

## RESULTS

### Size Frequency and Sex Ratios

From May 1988 to September 1989, a total of 870 *Panulirus guttatus* were caught, 558 of which were males (64.1%), and 312 of which were females (35.9%), i.e., the sex ratio was 1.8 males per female. Females were less abundant throughout the sampling period, especially in August 1988 and September 1989 (Fig. 1). The largest catches were obtained in May 1988 and April–May 1989, and the smallest in July and September 1988, and in February, August, and September 1989 (Fig. 1).

Females ranged in size from 35.2–74.9-mm CL (mean  $\pm$  SD =  $54.8 \pm 7.0$  mm), and males from 36.4–83.5-mm CL (mean  $\pm$  SD =  $61.1 \pm 9.5$  mm) (Fig. 2).

### Reproductive Size and Seasons

Of the 312 females, 125 (39.9%) were ovigerous. The size range of ovigerous females was 40.0–73.5-mm CL (Fig. 2), with a size at 50% maturity (in a cumulative curve of percentage of ovigerous females by size) of 56.4-mm CL. Ovigerous females were present every month, except during September of both 1988 and 1989 (Fig. 3), when the whole sample of lobsters was small (see Fig. 1). Ovigerous females were more prevalent

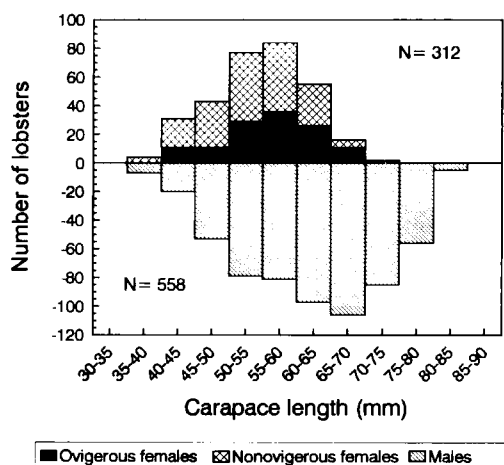


Fig. 2. Size distribution of male, ovigerous, and non-ovigerous female *Panulirus guttatus* in the total sample from May 1988 to September 1989.

in May 1988 (57.1%) and July 1989 (71.4%). When the data were grouped by calendar seasons, ovigerous females comprised 55.5% of the females in the population in the spring of 1988, 20% in the summer, 20.6% in the autumn, 32% in the winter, 48.6% in the spring of 1989, and 52.9% in the summer of 1989.

To determine whether there was a progression in peaks of ovigerous females by size class (Berry, 1971; Briones-Fourzán and Lozano-Alvarez, 1992), seasonal data were broken down in three classes: small (<50-mm CL), medium (50–60-mm CL) and large

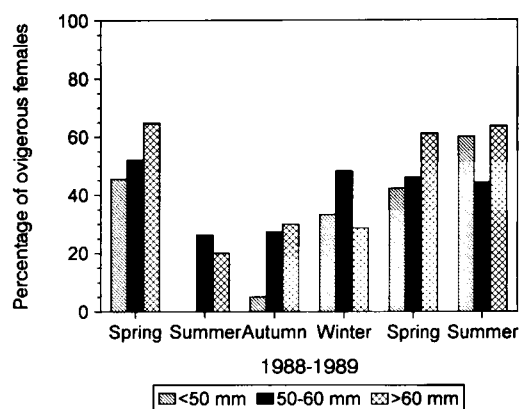


Fig. 4. Percentage of small (<50 mm carapace length, CL), medium (50–60 mm CL), and large (>60 mm CL) ovigerous female *Panulirus guttatus* in seasonal samples from May 1988 to September 1989.

(>60-mm CL) females. All three size classes, especially the small size, had low percentages of ovigerous females in the summer and autumn of 1988 (Fig. 4). About half of all medium-sized females were ovigerous during the spring of 1988, and the winter, spring, and autumn of 1989. In contrast, over 60% of large females were ovigerous in the spring of both years and in the summer of 1989. Small females showed high variability in reproduction: in the summer of 1987, no small females were ovigerous, whereas in the summer of 1988, 60% were in this condition.

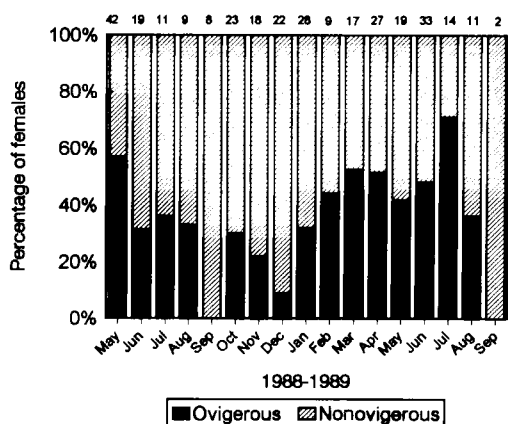


Fig. 3. Percentage of ovigerous and nonovigerous female *Panulirus guttatus* in monthly samples from May 1988 to September 1989. Numbers above bars indicate sample size.

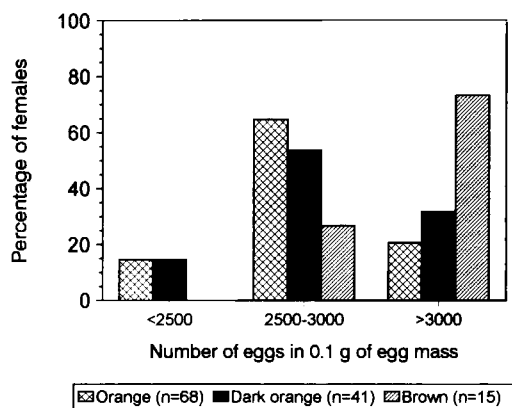


Fig. 5. Distribution of mean number of eggs in 0.1 g of egg mass (EM) in three stages of egg development, from female *Panulirus guttatus* sampled from May 1988 to September 1989 (orange eggs = newly spawned, dark orange eggs = intermediate development, brown eggs = about to hatch).

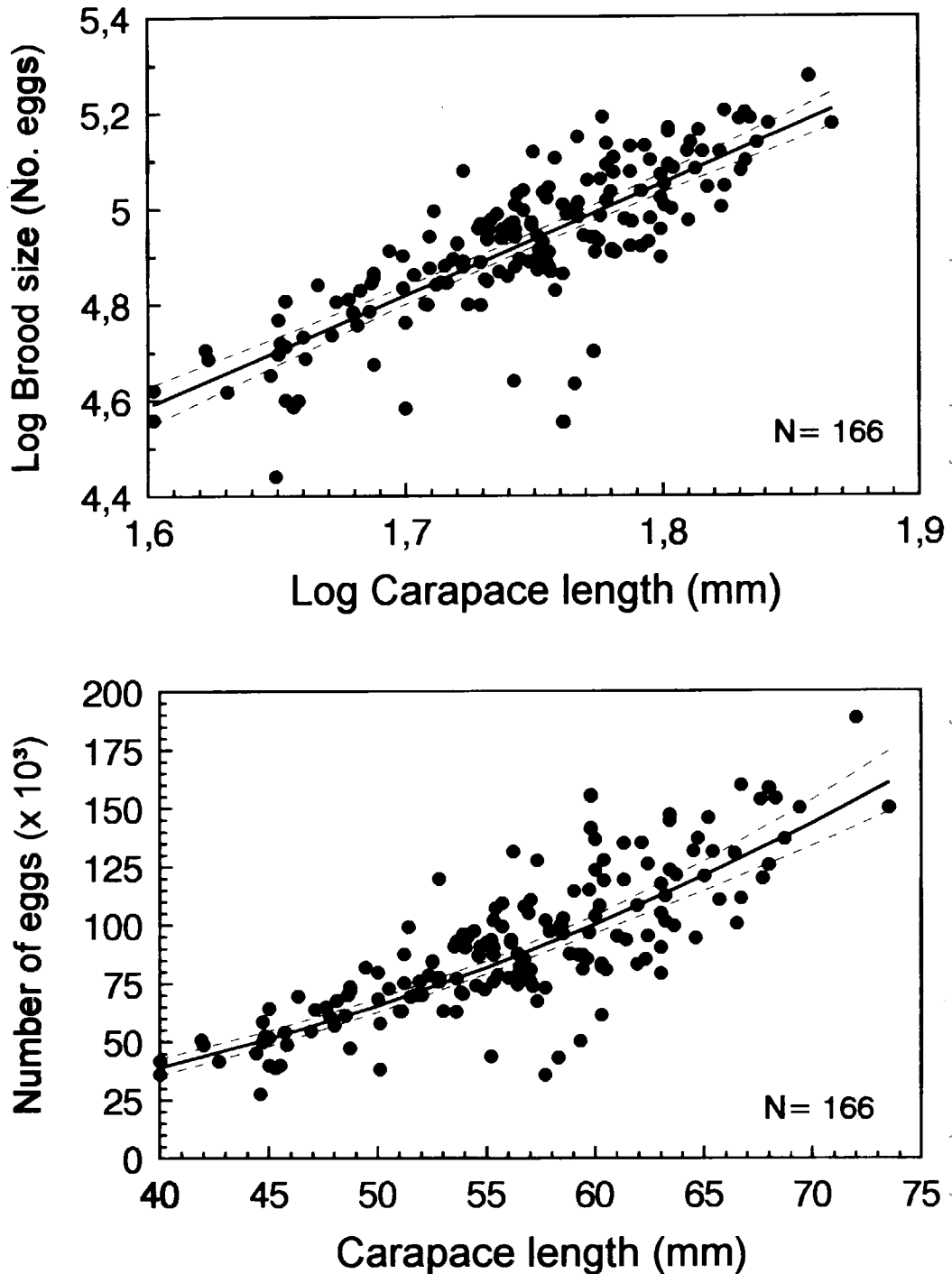


Fig. 6. Data plot, regression equations, and regression lines with 95% confidence limits for regression estimates for carapace length (CL) versus brood size (BS, number of eggs) relationship in female *Panulirus guttatus*, 1987-1989 (a) (top) with log-transformed data, (b) (bottom) with untransformed data. [Equation for (a):  $\text{Log BS} = 2.3567 (\text{log CL}) + 0.8134$ ; for (b):  $\text{BS} = 6.5073 \text{ CL}^{2.3567}$ ]

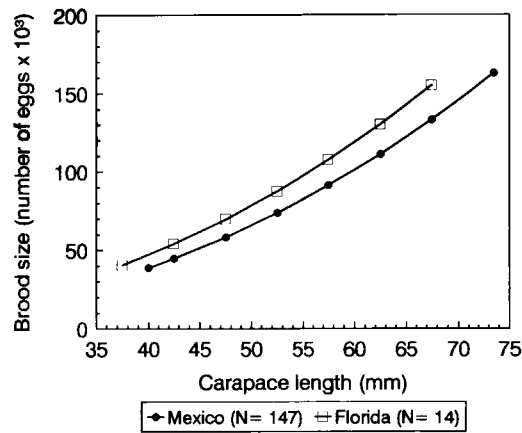


Fig. 7. Estimated mean brood size for female *Panulirus guttatus* from Mexico and Florida (equation and size range for Florida from Chitty, 1973).

Relative Egg Size

The range of the dry weight of egg masses was 0.93–7.22 g (mean ± SD = 3.25 ± 1.18 g). The number of eggs in 0.1 g EM ranged from 2,200–3,952 (mean ± SD = 2,846 ± 315). Over 50% of females with orange and dark orange eggs had between 2,500 and 3,000 eggs in 0.1 g EM (Fig. 5). In contrast, 73.3% of females with brown eggs had over 3,000 eggs in 0.1 g EM, indicating that eggs about to hatch were generally smaller, i.e., their average mass was lower than eggs in previous stages of development. After excluding the data of brown egg masses, no seasonal progression in egg size was apparent from the remaining data; most ovigerous females from the autumn of 1988 to the spring

of 1989, and all in the summer of 1989, had medium-sized eggs, suggesting that the size of the eggs remains fairly similar in subsequent spawnings.

Brood Size

The brood size of 159 ovigerous females ranged from 27,560 for a 44.6-mm CL female, to 188,440 eggs for a 72.0-mm CL female. However, because of the apparent reduction in size of eggs about to hatch, the data of females with brown eggs were not included in the BS–CL regressions. The regression coefficients of log BS versus log CL for the autumn of 1988, and the winter, spring, and summer of 1989 (Table 1a) were not significantly different ( $F = 1.404$ ,  $d.f. = 3, 68$ ,  $P > 0.25$ ), indicating that the mean brood size of female *P. guttatus* did not change as a function of season.

No significant differences were found between the regression coefficients of log BS versus log CL of the spring of 1988 and the spring of 1989 (Table 1a;  $t = 0.244$ ,  $d.f. = 55$ ,  $P > 0.5$ ), nor between those of the years 1987 and 1988–1989 (Table 1b;  $t = 0.627$ ,  $d.f. = 110$ ,  $P > 0.5$ ). Thus, a global regression equation of BS versus CL was computed with the whole data set (Table 1c). Figures 6a, b show the data plot, the estimated regression line with 95% confidence limits for the regression estimates, and the equations for the relationship BS versus CL, with the log-transformed data and the untransformed data, respectively.

The BS–CL relationship of *P. guttatus* has also been estimated from populations in Florida (Chitty, 1973) and Martinique (Marfin, 1978). However, Marfin employed a method

Table 1. Size range in carapace length (CL, mm), sample size, regression equations of log brood size (BS, number of eggs) versus log CL relationships, coefficients of determination ( $r^2$ ), and test of significance,  $F$ , for (a) seasonal, (b) annual, and (c) global samples of *Panulirus guttatus* from Puerto Morelos, Mexico (females with brown eggs, i.e., about to hatch, excluded).

| Period                  | Size range CL (mm) | N   | Regression equation              | $r^2$ | F      |
|-------------------------|--------------------|-----|----------------------------------|-------|--------|
| a) Seasonal             |                    |     |                                  |       |        |
| Spring 1988             | 41.9–68.7          | 28  | Log BS = 2.3339(log CL) + 0.8699 | 0.56  | 32.4*  |
| Autumn 1988             | 45.0–65.2          | 13  | Log BS = 1.9090(log CL) + 1.5524 | 0.70  | 25.5*  |
| Winter 1989             | 42.7–68.0          | 16  | Log BS = 2.8269(log CL) – 0.0045 | 0.73  | 53.9*  |
| Spring 1989             | 40.0–72.0          | 31  | Log BS = 2.2308(log CL) + 1.0826 | 0.87  | 261.9* |
| Summer 1989             | 46.9–68.0          | 16  | Log BS = 2.0799(log CL) + 1.2946 | 0.77  | 52.2*  |
| b) Annual               |                    |     |                                  |       |        |
| Total 1987              | 40.0–73.5          | 38  | Log BS = 2.4297(log CL) + 0.6605 | 0.73  | 67.1*  |
| Autumn 1988–Summer 1989 | 40.0–72.0          | 76  | Log BS = 2.2551(log CL) + 1.0041 | 0.73  | 239.6* |
| c) Global               |                    |     |                                  |       |        |
| 1987–1989               | 40.0–73.5          | 147 | Log BS = 2.3567(log CL) + 0.8134 | 0.68  | 302.9* |

\* $P < 0.005$ .

Table 2. Estimation of the Index of Reproductive Potential (IRP) for female *Panulirus guttatus* from Puerto Morelos, Mexico. A = proportion of females in a given size class over total females; B = proportion of ovigerous females in a given size class; C = average brood size of females in class; E = percentage of total egg production; F = E/A/100, i.e., an index of productivity of size class. (The value of D in the equation  $IRP = A \cdot B \cdot C / D$  equals 105.36.)

| Size class<br>(CL, mm) | Total number<br>of females | Number of<br>ovigerous females | A     | B     | C (×10 <sup>3</sup> ) | IRP   | Number of<br>eggs produced<br>per year (×10 <sup>3</sup> ) | E     | F    |
|------------------------|----------------------------|--------------------------------|-------|-------|-----------------------|-------|------------------------------------------------------------|-------|------|
| <40                    | 4                          | 0                              | 0.013 | 0.0   | 0.0                   | 0.0   | 0.0                                                        | 0.0   | 0.0  |
| 40–45                  | 31                         | 11                             | 0.099 | 0.355 | 44.8                  | 14.9  | 492.8                                                      | 4.5   | 0.45 |
| 45–50                  | 43                         | 11                             | 0.138 | 0.256 | 58.2                  | 19.5  | 640.2                                                      | 5.8   | 0.42 |
| 50–55                  | 77                         | 29                             | 0.247 | 0.377 | 73.7                  | 65.1  | 2,137.3                                                    | 19.3  | 0.78 |
| 55–60                  | 84                         | 36                             | 0.269 | 0.429 | 91.3                  | 100.0 | 3,286.8                                                    | 29.7  | 1.10 |
| 60–65                  | 55                         | 26                             | 0.176 | 0.473 | 111.1                 | 87.8  | 2,888.6                                                    | 26.1  | 1.48 |
| 65–70                  | 16                         | 11                             | 0.051 | 0.688 | 133.2                 | 44.4  | 1,465.2                                                    | 13.2  | 2.59 |
| >70                    | 2                          | 1                              | 0.006 | 0.500 | 157.6                 | 4.5   | 157.6                                                      | 1.4   | 2.33 |
| Total                  | 312                        | 125                            | 1.000 |       |                       |       | 11,068.5                                                   | 100.0 |      |

based on volume displacement, which appears to produce highly variable results (Chubb, 1994). Chitty (1973) used the gravimetric method similar to ours. Therefore, a comparison between the expected brood size using his log-log equation ( $\log BS = 2.28 (\log CL) + 1.02$ ) and ours (Table 1c) was performed (Fig. 7). For equivalent sizes of females, brood sizes in Florida were larger than in Mexico.

Index of Reproductive Potential

The constant D (105.36) in the IRP formula was computed to set the IRP of the 55–60-mm CL to 100 as a standard. Although *P. guttatus* is clearly capable of producing repetitive broods in view of its protracted spawning season and its short egg-incubation period (27–32 days; Chitty, 1973), there is no clear evidence of the number of broods that it can actually produce in one year, or the size of females able to spawn more than once. Therefore, we conservatively assumed one spawn per year. In general, the percentage of ovigerous females increased with size, from 35.5% in the size class 40–45-mm CL to 68.8% in the 65–70-mm class. Females in size classes 55–60- and 60–65-mm CL produced 55.8% of all eggs and had the highest IRPs, which were more reduced for smaller and larger females (Table 2). However, the 65–70-mm CL class had the highest productivity index ( $F = 2.59$ ), because of the prevalence of ovigerous females in this class and their larger brood sizes, making this size class 5.7 times more productive than the recently mature 40–45-mm CL class ( $F = 0.45$ ). This stresses the im-

portance of large females for the production of eggs in the population.

DISCUSSION

Sex ratios favoring males over females of *Panulirus guttatus* sampled with traps have been recorded in Bermuda (10:1, Sutcliffe, 1953), Jamaica (2.1:1, Munro, 1974), and in previous studies at Puerto Morelos (1.9:1, Carrasco-Zanini, 1985; 2.6:1, Negrete-Soto, 1988). In Martinique, tangle nets deployed directly on the coral reefs also caught more males than females (1.7:1, Farrugio, 1975; 1.9:1, Marfin, 1978). In contrast, sex ratios were more balanced in samples collected by diving in Florida (1.2:1, Caillouet *et al.*, 1971), or males were outnumbered by females (0.72:1, Chitty, 1973). Therefore, trapping and netting seem to provide samples with biased sex ratios.

Females with eggs in all developmental stages were found in our sampling locations on the coral reef, indicating that all reproductive activities, from mating to the hatching of eggs, may occur in the coral reef habitat (Briones-Fourzán and McWilliam, 1997). Owing to their protracted spawning season, which suggests repetitive breedings, it is possible that females are more sedentary than males, and/or that males are more aggressive in their search for food (Briones-Fourzán and Lozano-Alvarez, 1992). This would explain the higher proportions of males in traps and nets, which are caught during nightly movements, as opposed to daily catches by divers where sheltered lobsters have to be extracted from their dens (Morgan, 1974; Briones-

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Fourzán and Lozano-Alvarez, 1992). However, in samples of *P. guttatus* taken by divers in one locality in Florida, females were scarce from June–September, 1970 (Caillouet *et al.*, 1971), but predominated over males during the same period in 1971 (Chitty, 1973). Although *P. guttatus* is a typical reef-dwelling, nonmigratory palinurid (as opposed to migratory species, such as *P. argus*; Herrnkind, 1983), it is capable of short-range movements throughout its habitat in time, as shown by mark-recapture studies (Carrasco-Zanini, 1985; Negrete-Soto, 1988). Thus, an alternative explanation might be that males and females move differently throughout their habitat. This may be related to behavioral changes associated with reproductive activities or environmental factors that cause temporal changes in the catchability coefficient of each sex (Morgan, 1974).

Our smallest ovigerous female (40.0-mm CL) was larger than the 36.0-mm CL ovigerous female reported by Caillouet *et al.* (1971), and the 36.0-mm CL egg-bearing female found by Briones-Fourzán (1991) in the stomach of a large grouper (*Epinephelus* sp.). In contrast, the 73.5-mm CL ovigerous female caught in this study is the largest ever recorded for this species.

Ovigerous females of *P. guttatus* also occur throughout the year in Florida (Chitty, 1973). In contrast, egg-bearing females of *P. guttatus* have been reported in March–November in Jamaica (Munro, 1974), January–October in Martinique (Marfin, 1978), and August–October in Bermuda (Sutcliffe, 1953). In Puerto Morelos, variability in seasonal reproduction was high, but spring was an important reproductive season in both 1988 and 1989. Spring reproductive peaks occur in Florida (March–June; Chitty, 1973), and in Martinique (May–July; Farrugio, 1976; April–May; Marfin, 1978).

Chitty (1973) measured recently oviposited eggs of *P. guttatus* at 0.5–0.6 mm in diameter, but gave no figures for further stages. Freshly oviposited eggs of *Panulirus homarus* (L.) had a diameter of 0.54 mm, while eggs about to hatch measured 0.58 mm (Berry, 1971). Similarly, in *Panulirus inflatus* (Bouvier), eggs about to hatch were significantly larger (0.503 mm) than recently oviposited eggs (0.475 mm) (Gracia, 1985). It is possible that the reduction in weight (and therefore in size) of brown eggs in our samples

was due to a differential dehydration of eggs in distinct stages of development during the oven-drying process, rather than to an actual smaller size under natural conditions. An alternate explanation would be that development had consumed material and therefore brown eggs were of lower density. Therefore, this uncertainty warranted their exclusion from the estimation of the BS–CL relationships.

In contrast with the variability in the incidence of ovigerous females during the same season of different years, there were no significant seasonal or interannual variations in the BS–CL relationships. This was also reported for *P. argus* in a concurrent study conducted in the same area as ours (Fonseca-Larios and Briones-Fourzán, in press). The fact that this happened despite the passage of the great hurricane “Gilbert” on 14 September 1988, which caused severe and extensive damage to the coral reefs in the study area where *P. guttatus* dwells (Jordán-Dahlgren and Rodríguez, in press), suggests that the average brood size of tropical spiny lobsters in a given area is a feature rather resilient to natural catastrophic events. The large dispersion of points in the data plot (Fig. 6) suggests, however, that there might be a reduction in brood size at subsequent spawnings, as has been reported for *P. argus* (see Creaser, 1950).

The estimated IRPs indicate that, in this population of *P. guttatus*, females in the size classes of 55–65-mm CL produce over half of the total eggs and have similar indices of productivity. In addition, it is possible that females over the Lm of 56.4-mm CL spawn more than once a year, as has been shown in other tropical and subtropical palinurids (*P. homarus*, Berry, 1971; *P. argus*, Lipcius, 1985; *P. penicillatus* (Olivier), Juinio, 1987; *P. inflatus* and *P. gracilis* (Streets), Briones-Fourzán and Lozano-Alvarez, 1992). Thus, the relative importance of the largest size classes (55–60 mm and over) for egg production might be even higher than our estimates.

The difference in brood size at a similar CL estimated with Chitty’s (1973) equation and ours, does not necessarily imply that females of *P. guttatus* produce more eggs in Florida than in Mexico. Since Chitty’s sample size was small ( $N = 14$ ), it is possible that a larger sample from Florida would produce a result more similar to ours. However, Fonseca-Larios and Briones-Fourzán (in press) found dif-

ferences in brood size in female *P. argus* from locations in Mexico and Cuba at roughly the same latitude and a mere 350 km apart. They hypothesized that maturation processes and egg production of females from the same species may be related to the particular ecological characteristics and fishing pressure in different areas, regardless of their proximity. In the case of *P. guttatus*, this possibility needs to be pursued.

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## ANNOUNCEMENT

### 1998 Best Student Presentation Awards

The Crustacean Society recognized the following student presentations at the July, 1998 Summer Meeting held in Amsterdam, The Netherlands, in conjunction with the Fourth International Crustacean Congress. Awards were given for Best Student Paper (oral presentation) and Best Student Poster in each of five themes. Each of the ten award winners will receive a certificate, a free one-year membership in TCS (which includes a subscription to the Journal of Crustacean Biology), and a cash award of \$50.00.

#### Theme *Biodiversity*

Best Paper: Orpha Bellwood, James Cook University, Australia, "Evolution and Biogeography of the Calappidae (Decapoda: Brachyura)."

Best Poster: J. Olesen, Zoological Museum, Denmark, "Larval and Post-larval Development of the Clam Shrimp *Cyclestheria hislopi* (Branchiopoda, Conchostraca, Spinicaudata), with a Comparison of the Development in Various Cladoceran Orders."

#### Theme *Distributions*

Best Paper: S. M. (Roni) Robbins, The Natural History Museum, London, United Kingdom, "Seasonal Changes (1995–1997) in the Population Structure of Mitten Crabs (*Eriocheir sinensis*) in the Thames at Chelsea, London," (B. D. Smith, C. Byrne, P. S. Rainbow, and P. F. Clark, co-authors).

Best Poster: Sanja Gottstein, University of Zagreb, Croatia "Biodiversity of the Amphipod Fauna in Croatian Fresh and Brackish Water," (M. Kerovec, co-author).

#### Theme *Ecology and Behavior*

Best Paper: Katrin Berkenbusch, University of Otago, New Zealand, "Factors Influencing the Estimate of Sediment Turnover by the Burrowing Ghost Shrimp *Callinassa filholi* (Thalassinidae)," (A. A. Rowden, co-author).

Best Poster: Gauthier Chapelle, Institut Royal des Sciences Naturelles de Belgique, Brussels, Belgium, "Size in Amphipod Crustaceans: Polar Gigantism or Tropical Nanism?" (L. Peck, co-author).

#### Theme *Physiology and Biochemistry*

Best Paper: Silvia Gomez-Jimenez, University of Hull, United Kingdom, and Centro de Investigacion en Alimentacion y Desarrollo, Mexico, "The Effects of Hypo- and Hyper-Osmotic Media on Ammonia Efflux Rates and Haemolymph Consituent Levels in the Spiny Lobster, *Panulirus interruptus*," (R. F. Uglow and J. Hernandez-Lopez, co-authors).

Best Poster: Daniel Lemos, Universidade de São Paulo, Brazil, "Ontogenetic Variations of Digestive Proteinase Activity of Larval and Postlarval Shrimp *Penaeus paulensis*," (M. Hernandez-Cortes, M. Navarrete del Toro, F. Garcia-Carreño, and V. N. Phan, co-authors).

#### Theme *Fisheries and Aquaculture*

Best Paper: Kyra Schlining, Moss Landing Marine Laboratories, Monterey, California, USA; "The Spot Prawn, *Pandalus platyceros*, Trap Fishery in Carmel Submarine Canyon, California," (J. D. Spratt, co-author).

Best Poster: Eli Haugen, University of Bergen, Norway, "Parasites of Red King Crab, *Paralithodes camtschaticus*, from the Varangerfjord Area, Northern Norway," (G. A. Bristow & P. A. Jansen, co-authors).

In addition, honorable mention recognition was given to Vania Coelho from the Universidade de São Paulo, Brazil, for her talk entitled, "Comparison between the Setal Types Present on the Appendages Involved in Feeding of *Pomatogobia operculata* and *Upogobia omissa*," (S. de A. Rodriguez and A. B. Williams, co-authors).